Roll no.: 24BCB120D

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LAB SESSION 4: SEQUENCE DIAGRAM AND ACTIVITY DIAGRAM

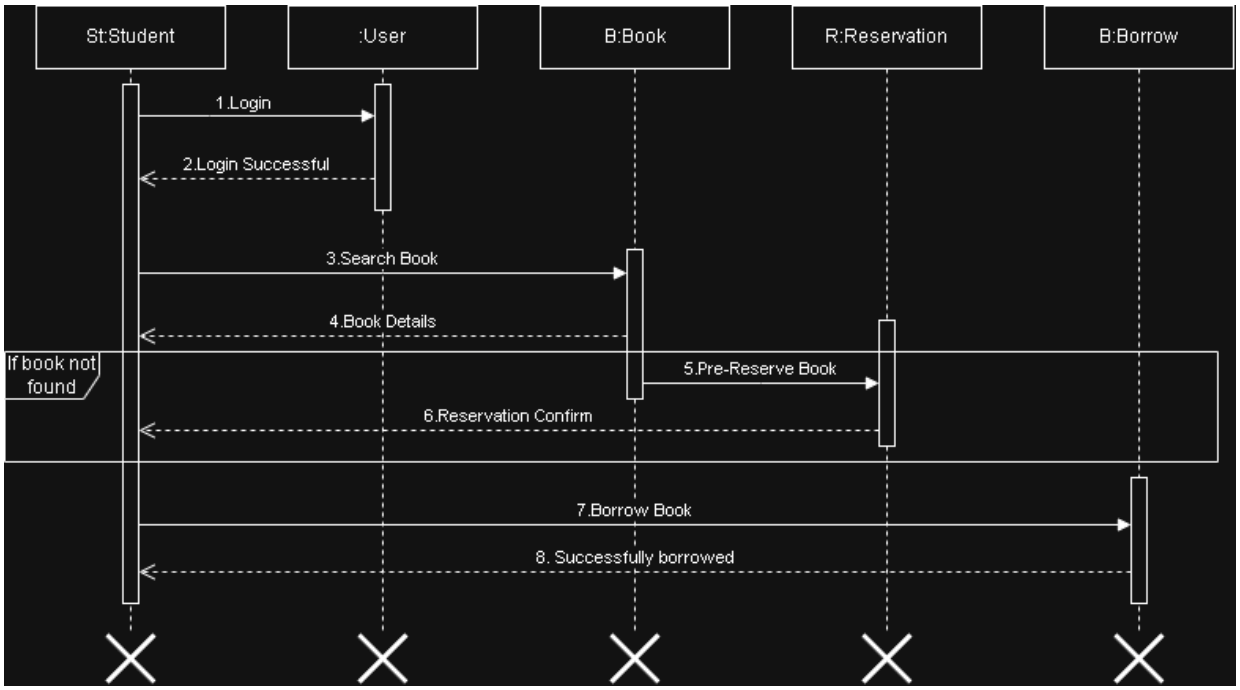
1. Sequence Diagram (Scenario: Borrow a Book)

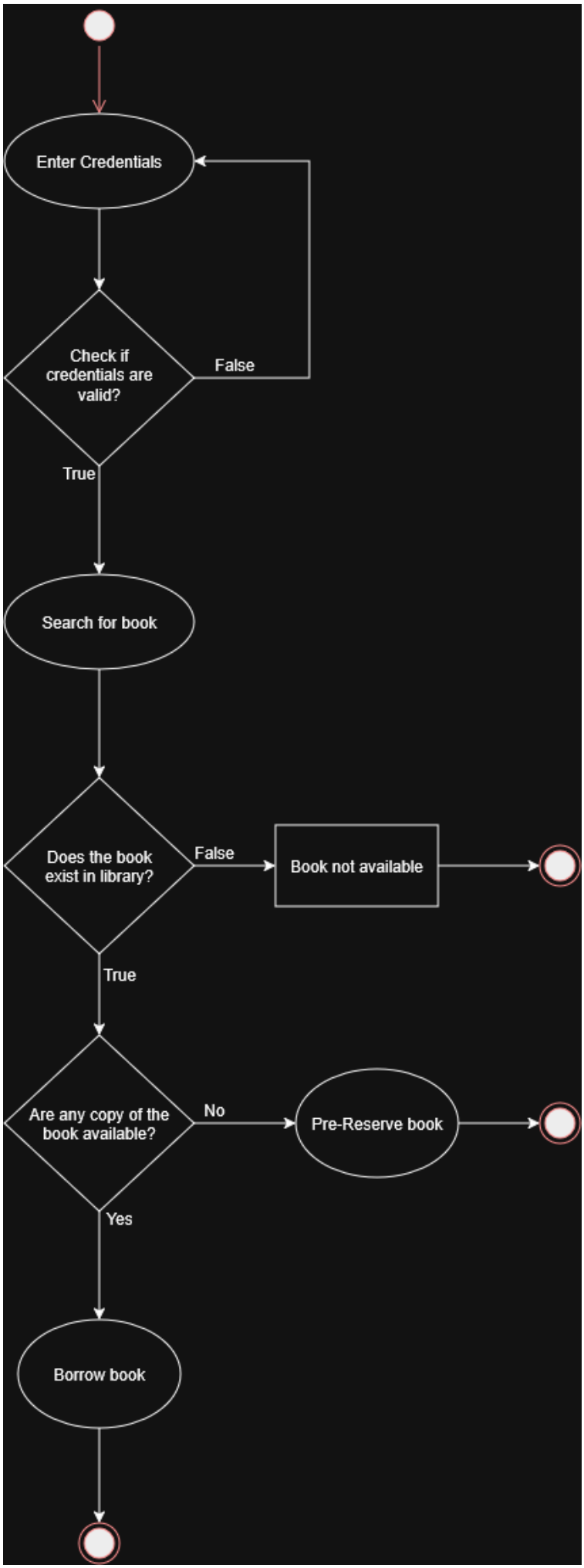
- Actor: Student

• Step 1: Student → User: Login.
• Step 2: User → Student: Login successful.
• Step 3: Student → Book: Search book.
• Step 4: Book → Student: Return book details.
• Step 5: Student → Reservation: Pre-reserve the book if no copy is available.
• Step 6: Reservation → Student: Reservation confirmed.
• Step 7: Student → Borrow: Borrow book.
• Step 8: Borrow → Student: Successfully borrowed.

Activity Diagram (Scenario: Borrow a Book)

- Start → Enter Credentials → Check if Credentials are Valid?
• False → Enter Credentials again.
• True → Search for Book → Does the Book Exist in Library?
  o No → Book Not Available → End.
  o Yes → Are Any Copies of the Book Available?
    o No → Pre-Reserve Book → End.
    o Yes → Borrow Book → End.




Enter Credentials
Check if credentials are valid?
False
True
Search for book
Does the book exist in library?
False
Book not available
True
Are any copy of the book available?
No
Pre-Reserve book
Yes
Borrow book

2. Sequence Diagram (Scenario: Renew a Book)

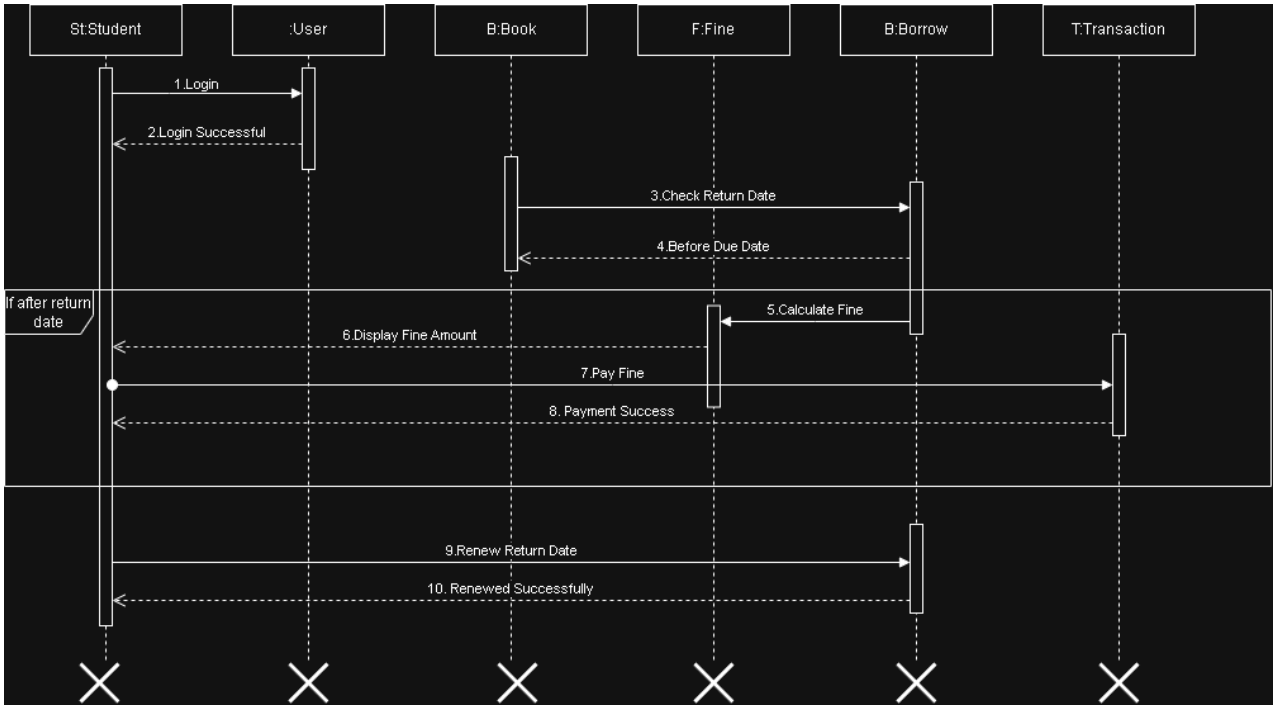
- Actor: Student

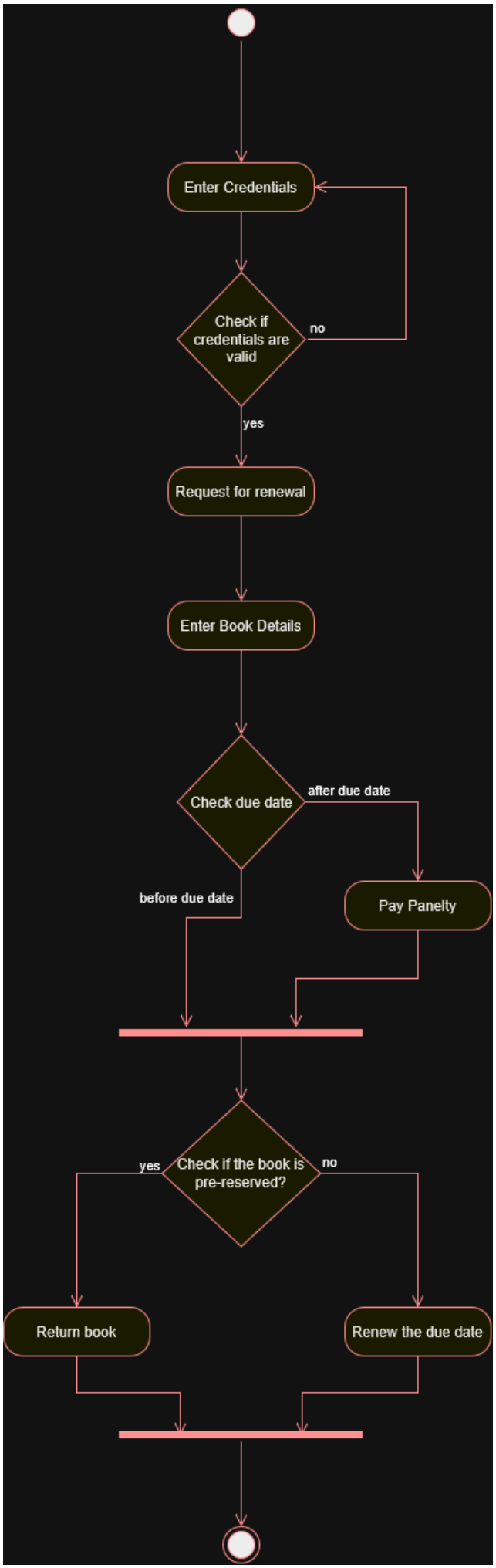
- **Step 1: Student → User: Login.**
- **Step 2: User → Student: Login successful.**
- **Step 3: Student → Borrow: Request to renew the book.**
- **Step 4: Borrow → Book: Check return/due date.**
- **Step 5: Book → Borrow: Return date information.**
- **Step 6: Borrow → Fine: Calculate fine if the book is overdue.**
- **Step 7: Fine → Student: Display fine amount.**
- **Step 8: Student → Transaction: Pay fine.**
- **Step 9: Transaction → Student: Payment successful.**
- **Step 10: Student → Borrow: Renew return date.**
- **Step 11: Borrow → Student: Return date renewed successfully.**

Activity Diagram (Scenario: Renew a Book)

- Start → Enter Credentials → Check if Credentials are Valid?

- **No → Enter Credentials again.**
- **Yes → Request for Renewal → Enter Book Details → Check Due Date.**
  - **Before due date → Continue.**
  - **After due date → Pay Penalty.**
- **Check if the Book is Pre-Reserved?**
  - **Yes → Return Book → End.**
  - **No → Renew the Due Date → End.**




Enter Credentials
Check if credentials are valid
no
yes
Request for renewal
Enter Book Details
Check due date
after due date
before due date
Pay Panelty
Check if the book is pre-reserved?
yes
no
Return book
Renew the due date

3. Sequence Diagram (Scenario: Return a Book)

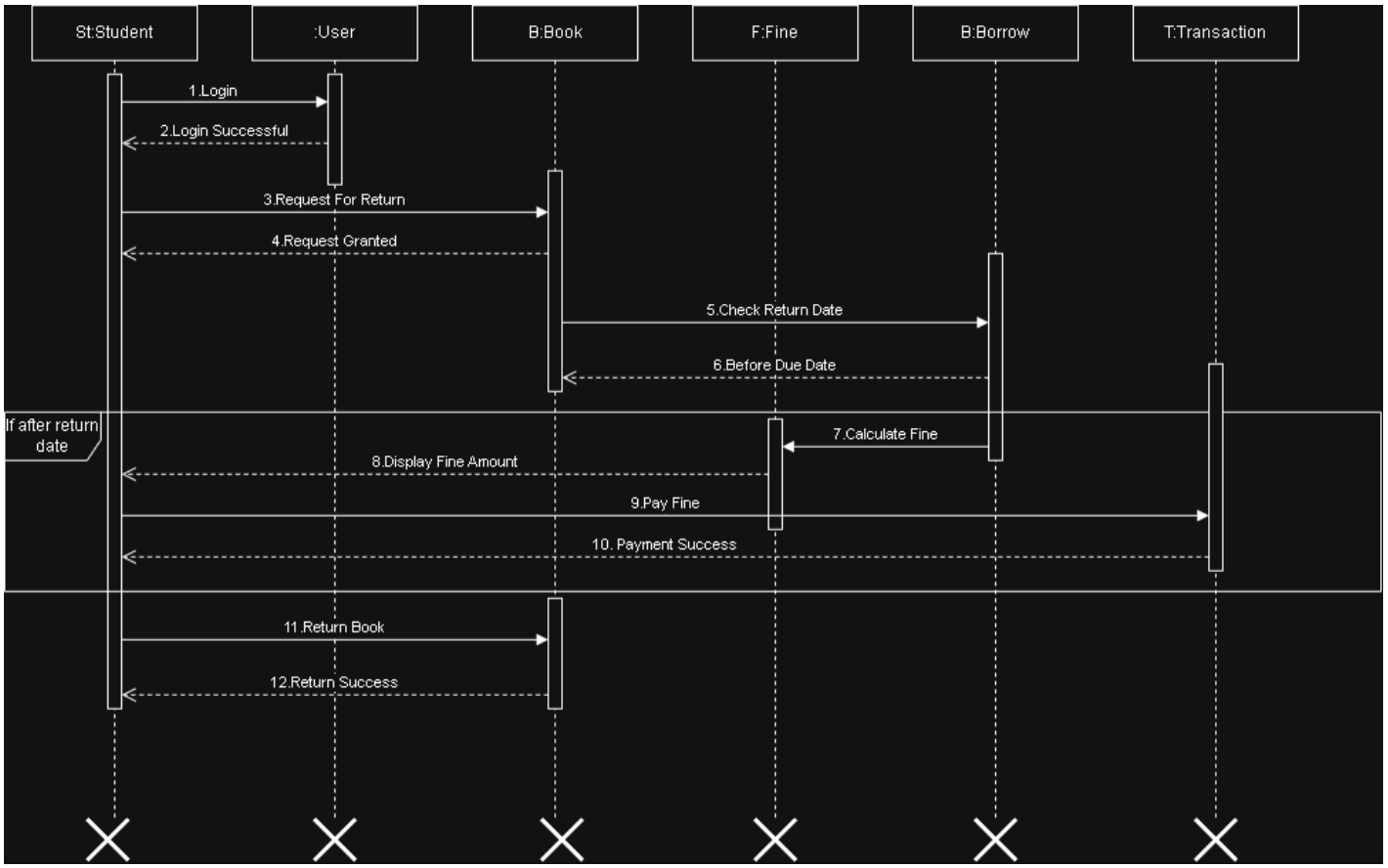
- Actor: Student

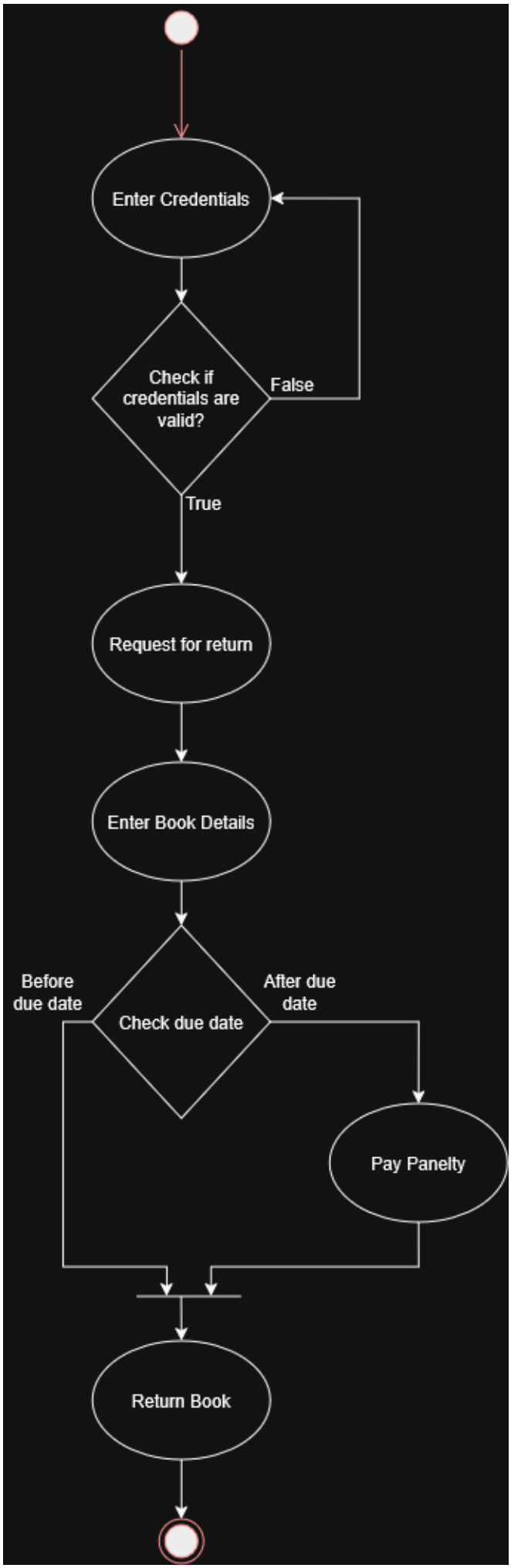
- **Step 1: Student → User: Login.**
- **Step 2: User → Student: Login successful.**
- **Step 3: Student → Borrow: Request to return the book.**
- **Step 4: Borrow → Borrow: Check return/due date.**
- **Step 5: Borrow → Student: Notify that the book is overdue.**
- **Step 6: Student → Fine: Pay fine.**
- **Step 7: Fine → Student: Payment successful.**
- **Step 8: Borrow → Book: Update status to "Available".**
- **Step 9: Book → Borrow: Status updated successfully.**
- **Step 10: Borrow → Student: Book returned successfully.**

Activity Diagram (Scenario: Return a Book)

- Start → Enter Credentials → Check if Credentials are Valid?

- **False → Enter Credentials again.**
- **True → Request for Return → Enter Book Details → Check Due Date.**
  - **Before due date → Return Book → End.**
  - **After due date → Pay Penalty → Return Book → End.**




Enter Credentials
Check if credentials are valid?
False
True
Request for return
Enter Book Details
Check due date
Before due date
After due date
Pay Panelty
Return Book